Customer Offer Engagement Dashboard – Technical Documentation

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Project Overview

Objective

The objective of this project is to analyze customer engagement and offer effectiveness using customer, offer, event, and transaction data. The dashboard provides business insights to support marketing and customer engagement decisions.

Tools Used

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Modeling

Dataset Information

Source Files

- customers.csv
- offers.csv
- events.csv
- data_dictionary.csv

Key Tables Created

Dimension Tables

- Customers
- Offers
- Offer_Channels

Fact Tables

- Offer_Received
- Offer_Viewed
- Offer_Completed

- Transactions

Data Model

Entity Relationship Diagram (ERD)

Insert Power BI Model View screenshot here.

Relationship Structure

- Customers → Offer_Received
- Customers → Offer_Viewed
- Customers → Offer_Completed
- Customers → Transactions
- Offers → Offer_Received
- Offers → Offer_Viewed
- Offers → Offer_Completed
- Offers → Offer_Channels

Data Preparation

Power Query Transformations

Customer Data

- Treated Age = 118 as missing values.
- Created Age Brackets for segmentation.
- Filled missing income values using median income by age bracket.
- Validated gender values.

Event Data

- Parsed JSON fields from the event value column.
- Extracted:
    - offer_id
    - reward
    - transaction_amount

Offer Data

- Validated reward values.
- Validated offer duration.

- Expanded communication channels into a separate Offer_Channels table.

JSON Extraction Process

The events dataset contained a column named **value**, where important information was stored in JSON format instead of separate columns.

Examples of information stored inside the JSON records included:

- offer_id
- reward
- transaction amount

To make the data suitable for analysis, Power Query was used to extract these values from the JSON structure and convert them into individual columns.

Extraction Steps

1. Identified JSON records stored in the value column.
2. Parsed the JSON data using Power Query.
3. Extracted key attributes such as:
    - offer_id
    - reward
    - transaction_amount
4. Converted extracted values into appropriate data types.
5. Validated the extracted values against sample records to ensure accuracy.

Business Benefit

Extracting JSON fields enabled the creation of relationships between tables and supported the calculation of key metrics such as:

- Offers Sent
- Offers Viewed
- Offers Completed
- Redemption Rate
- Average Transaction Amount

Without JSON extraction, the event data could not be effectively analyzed or used in dashboard visualizations.

Key DAX Measures

Offers Sent

COUNTROWS(Offer_Received)

Total Offers Completed

COUNTROWS(Offer_Completed)

Total Offers Viewed

COUNTROWS(Offer_Viewed)

Average Transaction Amount

AVERAGE(Transactions[transaction_amount])

Offer Redemption Rate

Customers Completed Offer ÷ Customers Received Offer

Completion Rate by Difficulty

Offers Completed ÷ Offers Sent

Dashboard Pages

Executive Overview

Provides business KPIs and summary metrics.

Offer Funnel Analysis

Measures conversion across the offer lifecycle.

Customer Segmentation

Analyzes engagement by demographic segments.

Offer Performance

Evaluates offer effectiveness by type, reward, channel, and difficulty.

Transaction Analysis

Compares spending patterns of completed and non-completed customers.

Key Findings

- Discount offers achieved the highest redemption rate.
- Reward value 4 generated the strongest customer response.
- Web channel delivered the highest redemption performance.
- Higher offer difficulty reduced completion rates.
- Customers who completed offers spent significantly more than non-completers.

Instructions for Adding New Offers

6. Add new offer records to the source dataset.
7. Verify offer attributes and channels.
8. Refresh Power BI Desktop.
9. Validate relationships and measures.
10. Publish updated reports if required.

Deployment

Power BI Service

1. Publish the .pbix file to Power BI Service.
2. Assign workspace permissions to authorized users.
3. Configure scheduled refresh.
4. Set refresh frequency to daily or weekly.
5. Monitor refresh status and resolve failures if necessary.